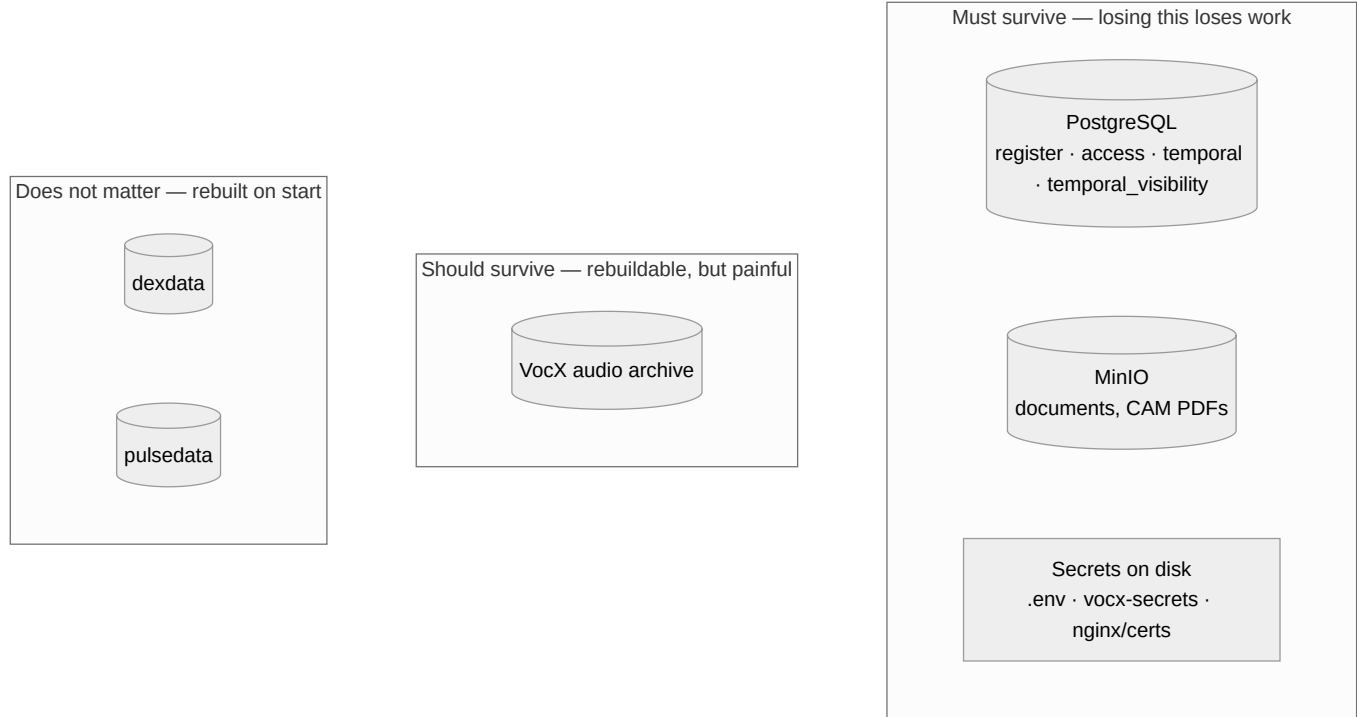


# 09 — Backup & Restore

**Audience:** whoever is accountable for not losing the book. **Companion docs:** [10 Upgrade & rollback](#) · [02 Deployment](#) · [13 Operations](#)

## 1. What has to survive



| Asset                                         | Where                         | Backed up by                                                          | Losing it means                                               |
|-----------------------------------------------|-------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------|
| PostgreSQL                                    | <code>pgdata</code> volume    | <code>pgbackup</code> nightly and <code>prism-deploy.sh backup</code> | Total data loss                                               |
| MinIO objects                                 | <code>miniodata</code> volume | <b>volume snapshot only — no automatic job</b>                        | Documents gone; rows still reference them                     |
| <code>deploy/compose/.env</code>              | bind mount                    | <code>prism-deploy.sh</code> secrets snapshot                         | Stack misconfigures; every secret lost                        |
| <code>deploy/vocx-secrets/</code>             | bind mount                    | secrets snapshot                                                      | Every user must re-authorise Google                           |
| <code>deploy/nginx/certs/</code>              | bind mount                    | secrets snapshot                                                      | The edge will not start                                       |
| VocX audio archive                            | volume                        | volume snapshot only                                                  | Recordings unrecoverable; transcripts survive in the Register |
| <code>dexdata</code> , <code>pulsedata</code> | volumes                       | —                                                                     | Rebuilt on next start                                         |

**The honest gap:** MinIO objects and the VocX audio archive have **no scheduled backup** today. The database does. If document bytes matter to you — and for a lender they do — add a volume snapshot or an `mc mirror` to off-box storage. \$7 gives both.

## 2. The three secret paths — say them out loud

```
deploy/compose/.env
deploy/vocx-secrets/
deploy/nginx/certs/
```

These are **never in git and never in a delivery archive**. They live only on the VM. That makes the secrets snapshot the *only* copy outside the running tree, and it makes verifying any zip you produce mandatory:

```
unzip -l "$ZIP" | grep -Ei "client_secret|token\.json|vocx-
secrets/*.json|vocx_tokens|compose/.env"
# exit code 1 (no matches) = clean. Any output at all = do not send that file.
```

## 3. The automatic database backup

`pgbackup` runs under the `backup` compose profile:

```
pgbackup:
  image: postgres:16-alpine
  profiles: ["backup"]
  environment:
    KEEP_DAYS: "${PGBACKUP_KEEP:-14}"
  volumes:
    - pgbackups:/backups
  command:
    - |
      while true; do
        ts=$(date +%Y%m%d-%H%M%S)
        pg_dumpall --clean --if-exists | gzip > "/backups/prism-$ts.sql.gz"
        find /backups -name 'prism-*.sql.gz' -mtime "+$KEEP_DAYS" -delete
        sleep 86400
      done
```

| Property  | Value                                                                |
|-----------|----------------------------------------------------------------------|
| Frequency | Every 24 h, counted from container start                             |
| Scope     | <code>pg_dumpall</code> — <b>all four databases</b> , roles included |
| Retention | 14 days ( <code>PGBACKUP_KEEP</code> )                               |
| Location  | The <code>pgbackups</code> volume, <b>on the same host</b>           |

Three things to be clear-eyed about:

1. **It is not a cron.** It is a `sleep 86400` loop, so a container restart moves the backup time. Do not expect it at a fixed hour.

2. **It stays on the same machine.** A host loss loses the backups with the data. Copy them off-box (§7).
3. **Nothing verifies restorability.** A dump that exists is not a dump that restores. Do the drill in §6.

```
# Confirm it is running and see what it has
docker compose -f deploy/compose/docker-compose.yml --profile backup ps pgbackup
docker compose -f deploy/compose/docker-compose.yml run --rm -v compose_pgbackups:/b \
  postgres:16-alpine ls -lh /b
```

## 4. The deploy-script backup (the one taken before every upgrade)

```
./prism-deploy.sh backup      # a dump + a secrets snapshot, and nothing else
```

This runs automatically as the **first step** of every `upgrade`. Rule 1 of the script: *"NOTHING IS TOUCHED UNTIL A BACKUP EXISTS."*

### The database dump — and how it refuses to lie to you

```
dc "$LIVE" exec -T postgres pg_dumpall -U prism | gzip > "$dump"
gzip -t "$dump"                                || die "the dump is corrupt (gzip -t failed)"
(( bytes > 100000 ))                            || die "the dump is only ${bytes} bytes – refusing to treat
that as a backup"
```

Three checks, each closing a real failure: postgres not running at all, a truncated gzip stream, and a dump that "succeeded" but is empty. **A backup that is not verified is not a backup**, and this is what that principle looks like in code.

### The secrets snapshot

```
SECRET_PATHS=(deploy/compose/.env deploy/vocx-secrets deploy/nginx/certs)
tar -czf "$tarball" -C "$LIVE" "${present[@]}"
chmod 600 "$tarball"
```

If the tar fails — most commonly a root-owned `nginx/certs/tls.key` while running as `ubuntu` — the script stops and names the file and both remedies. It does **not** proceed with a partial snapshot.

### Where backups land

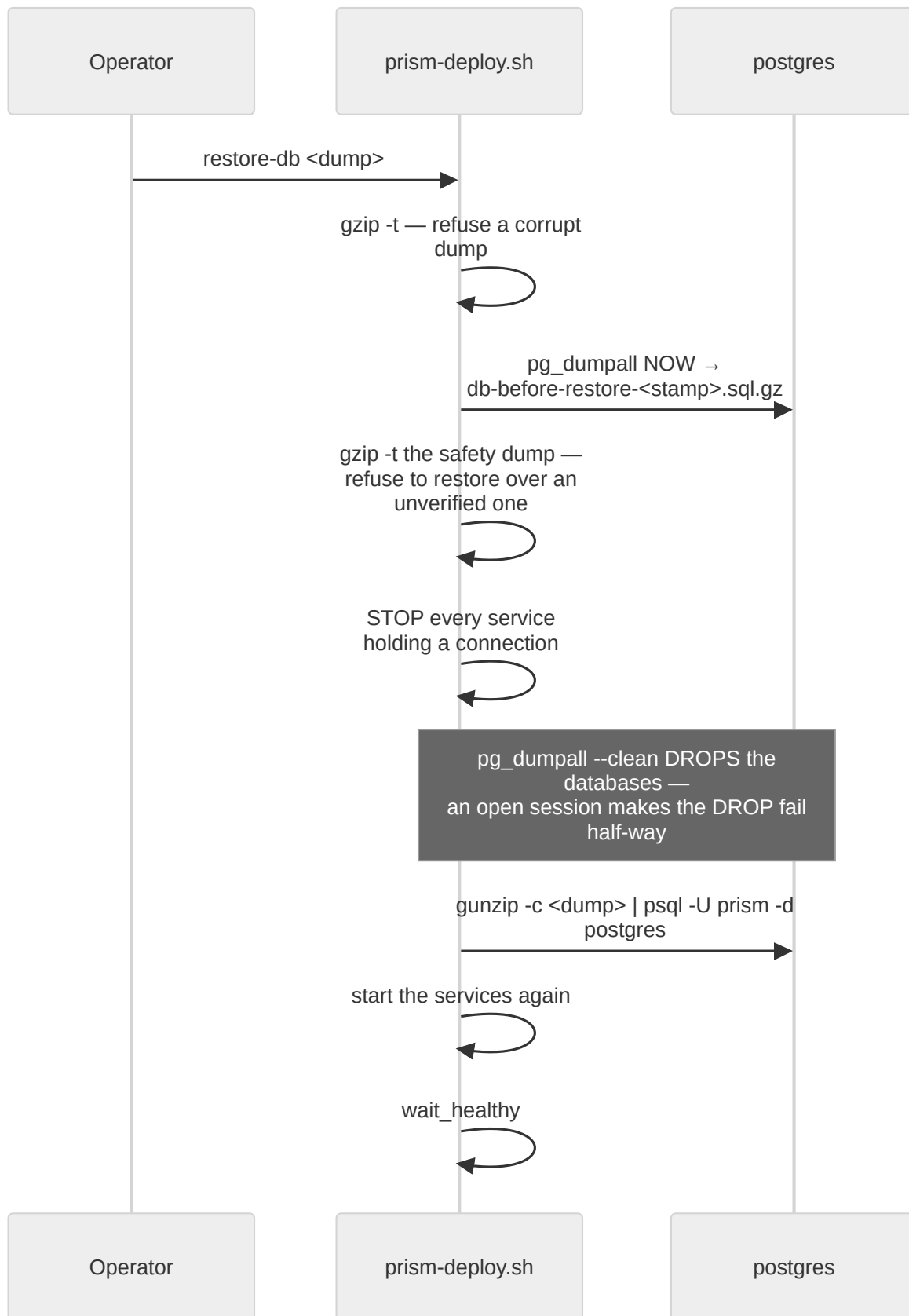
```
$PRISM_ROOT/backups/
├─ db-20260819-041500.sql.gz          ← pre-upgrade dump
├─ secrets-20260819-041500.tar.gz     ← chmod 600
├─ images-20260819-041500.tsv        ← the rollback image map
├─ db-before-restore-<stamp>.sql.gz   ← safety dump taken before any restore
└─ deploy-20260819-041500.log
```

Retention: `PRISM_KEEP_BACKUPS` (default 10).

## 5. Restoring

### The full-database restore

```
./prism-deploy.sh restore-db ~/aug_11/backups/db-20260819-041500.sql.gz
```



Two design points worth keeping:

- **A restore replaces, so the present is captured first.** The step is itself reversible.
- **Connection holders are stopped explicitly** — `register access gateway atlas vocx pulse workflows notifier orchestrator temporal temporal-ui`. A half-applied `--clean` restore is a far worse state than a refused one.

## Restoring by hand

```
cd ~/aug_11/prism
docker compose -f deploy/compose/docker-compose.yml stop \
  register access gateway atlas vocx pulse workflows notifier orchestrator temporal

gunzip -c /path/to/prism-20260819.sql.gz | \
  docker compose -f deploy/compose/docker-compose.yml exec -T postgres psql -U prism -d postgres

docker compose -f deploy/compose/docker-compose.yml start \
  register access gateway atlas vocx pulse workflows notifier orchestrator temporal
docker compose -f deploy/compose/docker-compose.yml exec nginx nginx -s reload
```

That last line matters — see [13 Operations](#) on stale upstream IPs.

## Restoring the secrets

```
tar -xzf ~/aug_11/backups/secrets-20260819-041500.tar.gz -C ~/aug_11/prism
ls -l ~/aug_11/prism/deploy/compose/.env \
  ~/aug_11/prism/deploy/vocx-secrets \
  ~/aug_11/prism/deploy/nginx/certs
```

`upgrade` does this automatically into the new tree, and **hard-fails** if `.env` is absent afterwards.

## 6. The restore drill — do this quarterly

An untested backup is a hope. The drill takes about twenty minutes and is the only thing that converts it into a recovery position.

```
# 1. On a scratch machine (or a second compose project name), bring up postgres alone
docker run -d --name drill -e POSTGRES_USER=prism -e POSTGRES_PASSWORD=x postgres:16-alpine

# 2. Restore the newest dump
gunzip -c prism-20260819-041500.sql.gz | docker exec -i drill psql -U prism -d postgres

# 3. Prove the book is there
docker exec -i drill psql -U prism -d register -c \
  "select count(*) from entities; select count(*) from deals;
  select count(*) from lending_tracker; select count(*) from interactions;"

# 4. Prove identity came too
docker exec -i drill psql -U prism -d access -c \
  "select count(*) from users; select count(*) from access_grants;"

# 5. Tear down
docker rm -f drill
```

Record the row counts. A dump whose counts have quietly collapsed is the failure you are drilling for.

## 7. Getting backups off the box (recommended, not yet built)

Everything above keeps backups on the same host. Two additions close that:

```
# Database dumps → S3, nightly
aws s3 sync ~/aug_11/backups/ s3://evam-prism-backups/${hostname}/ \
  --exclude '*' --include 'db-*.sql.gz' --include 'secrets-*.tar.gz'

# MinIO objects → S3 (or another MinIO), nightly
docker compose -f deploy/compose/docker-compose.yml exec -T minio \
  mc mirror --overwrite local/prism s3remote/evam-prism-documents
```

If you add the secrets tarball to an off-box location, that location now holds every secret PRISM has. Encrypt it and restrict who can read it — this is not a place to be casual.

## 8. RPO and RTO, honestly stated

|                                | Today                                                                                                  | With off-box sync                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>RPO</b> (data you can lose) | Up to <b>24 h</b> — the gap between nightly dumps. A pre-upgrade dump narrows it only around upgrades. | Same 24 h, but survives host loss |
| <b>RTO</b> (time to recover)   | ~15–30 min from a local dump: restore, start, verify                                                   | Add transfer time                 |
| <b>Host loss</b>               | <b>Total loss</b> — backups are on the same volume                                                     | Recoverable                       |
| <b>Document bytes</b>          | <b>No backup</b> — MinIO is unprotected                                                                | Recoverable                       |

If a 24-hour RPO is not acceptable for the book, the honest fix is WAL archiving ( `archive_mode=on` plus `pg_receivewal` or a base-backup tool) rather than more frequent `pg_dumpall`. That is a real change and it is not in place today.

## 9. What is *not* backed up, and why it is usually fine

| Not backed up          | Why acceptable                                                                          | When it is not                             |
|------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------|
| Temporal history       | It rides inside <code>pgdata</code> , so <code>pg_dumpall</code> <b>does</b> capture it | —                                          |
| <code>dexdata</code>   | Dex re-seeds on start; identities live upstream                                         | If Dex is your only user store             |
| <code>pulsedata</code> | Rebuilt by the next scan                                                                | Never                                      |
| Container images       | Rebuildable from source; <code>prism-deploy.sh</code> also tags them for rollback       | If the source archive is lost too          |
| VocX audio             | Transcripts and structured data are in the Register                                     | If recordings are evidence you must retain |

That last row is a policy question, not a technical one. If audio is retained evidence, back the volume up.

## 10. Checklist

### Weekly

- ☐ `docker compose --profile backup ps pgbackup` — still running?
- ☐ Newest dump is < 24 h old and > 100 KB
- ☐ Disk headroom on the backup volume

### Quarterly

- ☐ Run the restore drill (§6) and record the row counts
- ☐ Confirm the secrets tarball opens and contains all three paths
- ☐ Confirm the TLS certificate in the snapshot is the current one

### Before anything risky

- ☐ `./prism-deploy.sh backup`
- ☐ Note the dump path — `rollback --with-db` and `restore-db` both need it